



GITHUB COPILOT

Your Team's New Teammate

GitHub Copilot Across the Delivery Lifecycle

45 Minutes

Product Owners

Project Managers

Engineering Leaders

The Four Questions This Talk Answers



Why should my team use this?

Evidence-backed business case



Why should I encourage adoption?

Competitive reality & team health



What features should we encourage?

Practical adoption roadmap



How do I measure the value?

Metrics framework & ROI



The Business Case

Evidence, competitive context, and what it means for your team



The Window for "Wait and See" Has Closed

15M+

Active Users

Globally, and growing

90%

Fortune 100

Already deployed

50K+

Organizations

Across every industry



What is Copilot, really?

Not just autocomplete. An AI system that writes code, reviews PRs, generates tests, documents changes, and — increasingly — autonomously completes tasks from your backlog.



What does "Agentic" mean?

Assign a GitHub Issue → Copilot creates a branch, writes the code, runs tests, opens a PR. A teammate who works while others sleep, never misses a standup, and always follows your standards.



The Research Is In — Fresh Evidence From 2025–2026

GitHub Octoverse 2025 — Platform Telemetry (180M+ devs)

Coding Agent rolled out in May 2025 · Sept 2024–Aug 2025 data window

1M+

PRs from Coding Agent in 4 months

Octoverse 2025 — Platform-Wide Throughput

43.2M PRs merged/month avg · 986M commits in 2025

+23%

PR throughput year over year

GitHub Code Review Study (mid-2025)

In-depth interviews with developers using Copilot code review

72.6%

report improved effectiveness

Echoes of AI — Independent RCT (Borg et al. 2025)

N=151 professional developers · GitHub Copilot + Cursor · Java tasks

30.7%

median time reduction on dev tasks

Microsoft Research, SPACE of AI (Aug 2025, N>500): efficiency & satisfaction up across teams · benefits scale with task routineness, not raw seniority



What This Means for Your Delivery Cycle

✗ Before Copilot

PR reviews: 3+ day wait

Senior devs are the bottleneck

Boilerplate eats sprint capacity

Test writing, docs, repetitive setup

New devs slow for months

Learning codebase = high PM overhead

Stories carry over every sprint

Velocity unpredictable

✓ After Copilot

PR reviews: same-day

AI reviews first, humans approve

Boilerplate is automatic

Devs focus on the hard, interesting work

New devs ramp in weeks

Copilot is their knowledgeable pair programmer

Sprint velocity stabilizes

More predictable planning and commitments



How Copilot Works

Across every phase of your delivery lifecycle



Copilot Touches Every Phase of the SDLC



Requirements

Translates stories to tasks
Flags missing edge cases
Reduces mid-sprint surprises



Development

Writes code from descriptions
Follows your patterns
Agent picks up issues autonomously



Code Review

Every PR reviewed instantly
Security + quality + coverage
3 days → same day



Testing

Generates test suites
Edge cases covered
Fewer bugs escaping to QA



Ops & Maintenance

Writes changelogs & runbooks
Triage user issues
Patches dependencies

The Agentic flywheel: PO writes clear story → Agent codes & tests → Code Review catches issues → Developer approves → Copilot documents → PO validates ✓

⚡ The Coding Agent: Assign Issues to Copilot

How It Works

- 1 Assign a GitHub Issue to Copilot**
From the issue page, assignees dropdown, or Copilot Chat
- 2 Agent starts working**
Runs in a secure, sandboxed environment. Creates a branch.
- 3 Copilot writes, tests, and proposes**
Opens a pull request when complete. No auto-merge.
- 4 Developer reviews & approves**
Human owns every merge. Full audit trail.

Best Suited For

- ✓ Bug fixes with clear reproduction steps
- ✓ Adding test coverage to existing code
- ✓ Documentation updates and changelogs
- ✓ Well-specified incremental features
- ✓ Dependency updates and security patches
- ⚠ Vague stories with unclear acceptance criteria
- ⚠ Complex multi-system architectural changes

PO insight: The quality of your acceptance criteria is now the direct input to agent performance.

Features to Encourage — Phased Adoption Roadmap

TIER 1 · Days 1–30

Copilot Chat

Q&A, code explanations, brainstorming, drafting. Replaces "quick question" Slack interruptions.

Copilot Code Review

Automatic PR review for security, quality, coverage. PR queue drops to same-day.

Target: 80% weekly active use

TIER 2 · Days 30–60

Coding Agent

Assign GitHub Issues to Copilot. Low-medium complexity tasks completed autonomously.

Custom Instructions

Team standards baked in. Consistent quality without repeated review feedback.

Target: 1+ agent task/sprint

TIER 3 · Days 60–90

MCP Integrations

Connect Copilot to Jira, Confluence, Azure DevOps. AI sees your actual requirements.

Multi-Agent Workflows

Multiple agents collaborating on a larger feature. For teams with mature AI practices.

Target: Workflow integration



Measuring Impact

Metrics, DORA, and your 90-day measurement plan



Measuring Value: The GitHub Metrics Dashboard

Built-In Metrics (No Setup Required)



Active Users

Daily / weekly — who's using it consistently



Acceptance Rate

>25% = suggestions are relevant and useful



Feature Adoption Breakdown

Chat vs. Code Review vs. Coding Agent



AI Adoption Cohorts

Track teams on their maturity journey



REST API

Pipe data into your own dashboards

Connect to DORA Metrics

Deployment Frequency

More throughput → more frequent, smaller releases

Lead Time for Changes

Faster coding + review → shorter commit-to-deploy

Change Failure Rate

Better coverage + review → fewer bad deploys

Mean Time to Restore

Faster diagnosis + fix generation → quicker recovery

Track PR cycle time (open → merge) as your leading indicator



Your 90-Day Measurement Plan



Weeks 1–4: Baseline

Pull DORA metrics from GitHub Insights

Measure PR cycle time (open → merge)

Survey: hours/week on mechanical work?

Record sprint carry-over rate



Weeks 5–8: Track Adoption

Weekly active users in dashboard

Feature breakdown: what's being used?

Flag teams <50% for coaching

First Coding Agent task?



Weeks 9–12: Measure Impact

Compare PR cycle time: before vs. after

Sprint carry-over rate: changed?

Re-survey developer satisfaction

Calculate ROI

Simple ROI: (10 devs × 5 hrs/week saved × \$75/hr × 52 weeks) = **\$195,000/year** vs. **\$2,280/year** license cost → **~85x ROI**

The Human in the Loop

What AI Does Better

- ⚡ Execute well-specified tasks at speed
- ⚡ Review code consistently at any hour
- ⚡ Generate boilerplate without fatigue
- ⚡ Follow standards without reminders
- ⚡ Work in parallel across multiple tasks

What Stays Irreversibly Human

- 🎯 **Product judgment** — what should we build?
- 💛 **Stakeholder trust** — relationships & alignment
- ⚖️ **Ethical assessment** — is this the right thing?
- 👂 **User empathy** — does this solve the real problem?
- 🏆 **Final validation** — acceptance criteria sign-off

Your new superpower as a PO: Well-written stories with clear acceptance criteria now have a direct multiplier on delivery speed. Invest in story quality.



Your Action Plan

Three horizons: this week, this month, this quarter

→ Your Next Steps



This Week

- ☐ Ask: does your team have Copilot licenses?
- ☐ Ask: are they using them consistently?
- ☐ Identify the access or adoption gap



This Month

- ☐ Enable the Copilot Metrics Dashboard
- ☐ Establish your DORA baseline
- ☐ Drive Tier 1 adoption to >80% weekly



This Quarter

- ☐ Set sprint velocity improvement target (15–25%)
- ☐ Trial Coding Agent on 2–3 items/sprint
- ☐ Calculate ROI with the 90-day framework

The competitive risk of inaction: Octoverse 2025 logged 1M+ Copilot Coding Agent PRs in 4 months and +23% YoY platform-wide PR throughput. Every sprint without adoption is a sprint where the gap widens.

Sources & Further Reading — 2025–2026 Evidence

GitHub Octoverse 2025

1M+ Coding Agent PRs in 4 months · +23% YoY PR throughput
github.blog/news-insights/octoverse ↗

Copilot Coding Agent — GA Launch (May 2025)

Carvana, EY, Avanade enterprise rollouts
github.blog/news-insights/product-news ↗

GitHub Code Review Study (mid-2025)

72.6% report effectiveness gains · ~33% less PR back-and-forth
github.blog/ai-and-ml ↗

GitHub Copilot Usage Metrics Docs

Dashboard guide & REST API reference
docs.github.com/en/copilot ↗

Microsoft Research — SPACE of AI (Aug 2025)

N>500 developers · efficiency & satisfaction up; benefits heterogeneous
arxiv.org/abs/2508.00178 ↗

Echoes of AI — Two-Phase RCT (Borg et al. 2025)

N=151 professionals · 30.7% time reduction · no maintainability harm
arxiv.org/abs/2507.00788 ↗

METR — Counter-Evidence RCT (July 2025)

Senior OSS devs 19% slower on familiar code · context matters
arxiv.org/abs/2507.09089 ↗

DORA Research

Gold-standard DevOps delivery metrics framework
devops-research.com ↗

Full reference list at exec-talks/copilot-sdlc-pm/README.md